AGENTIC AI SUMMIT 2026 · UC BERKELEY · AUG 1, 2026

Are Small LLMs Ready for Coding Agents?

A tight harness connects controllable models to accepted full workflows.

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WHY THIS MATTERS NOW

A tight loop connects model control to full workflows

The model gets one bounded move; the harness keeps the end-to-end contract.

01

CONTROLLABLE SMALL LLMS

Choose the model boundary

- Execution speed
- Hardware and deployment
- Post-training access

→

02

ASKME LOOP

One small action at a time

```
plan → act → execute / test
→ fresh evidence ↺
```

Interpret the result. Update locally when possible.

→

03

ACCEPTED FULL WORKFLOW

Keep and check the contract

- Preserve completed work
- Check required behavior
- Accept the real artifact independently

The bridge: small action surface + fresh execution feedback + independent workflow acceptance.

A RETAINED INTEGRATION MISS

The code ran. The workflow still failed.

REQUIRED CONTRACT

```
write msg.h + main.c
build ./main
run ./main
observe REPLAN_OK
```

Source and header written

The requested program was correct.

`/tmp/test` compiled

A different output path was chosen.

`/tmp/test` ran successfully

It printed `REPLAN_OK`.

Agent reported completion

The independent acceptance test found no `./main`.

Every recorded action succeeded. The deliverable contract still drifted.

One hosted build cell · retained under the predeclared failure rule · evals/draft-results.json

DESIGN HYPOTHESIS

Reason over fresh evidence. Preserve progress.

01
Keep the contract
Carry the required behavior and artifact forward.

→

02
Choose one action
Advance the current plan with bounded scope.

→

03
Execute
Run the tool, focused test, or integration check.

→

04
Interpret
Use the new evidence, not reconstructed state.

→

05
Update locally
Change only what the evidence invalidated.

Expected result
Continue to the next planned action.

Local mismatch
Repair the affected step and rerun its check.

Broken assumption
Replan broadly only when the plan is no longer valid.

Trajectory goal: fewer repeated failures, fewer stuck steps, and less unnecessary plan churn—not longer monologues.

MEASURED HARNESS SMOKE · JULY 10, 2026

Four models, eight descriptive receipts

Two variants per family; architecture and active parameters changed.

8 / 8 agent complete

7 / 8 artifact accepted

Hosted · simple checks · n=1/cell · descriptive only

MODEL	SHAPE	BUILD OBSERVATION	REPAIR OBSERVATION
Gemma 4 26B A4B	◆ MoE 25.2B total · 3.8B active	ACCEPTED 603.6s	ACCEPTED 20.0s
Gemma 4 31B	● Dense 30.7B parameters	ACCEPTED 66.5s	ACCEPTED 22.4s
Qwen3.6-27B	● Dense 27B parameters	ACCEPTED 47.9s	ACCEPTED 23.0s
Qwen3.6-35B-A3B	◆ MoE 3B active	NOT ACCEPTED wrong artifact path · 17.7s	ACCEPTED 11.8s

Boundary: architecture, active parameters, run order, and trajectories changed; the 31B follow-up was post-hoc. No size, family, reasoning, or reliability inference.

PRE-AUGUST NATIVE PILOT

Freeze the smallest clean test

01

4 frozen workflows

Valid multi-file code, semantic failures, visible feedback, held-out scoring.

02

2 frozen policies

Explicit-reasoning off versus the current composite gated policy.

03

3 randomized repeats

$4 \times 2 \times 3 = 24$ scheduled runs per model.

04

2 primary outcomes

Held-out acceptance and false completion across all valid runs.

Secondary: repeated actions · work redone · local/full replans · latency · tokens

Publish tasks, policies, budgets, gate table, randomization, and exclusions before the first outcome-bearing call.

ANSWER TO THE TITLE

Promising—with a tight harness.

- Model choice
- Simple contract
- Execution / test
- Local reasoning
- Accepted workflow

Make the interface easier to use.
Keep success tied to real behavior.

The smoke validates this interface; readiness needs repeated held-out workflows.

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data